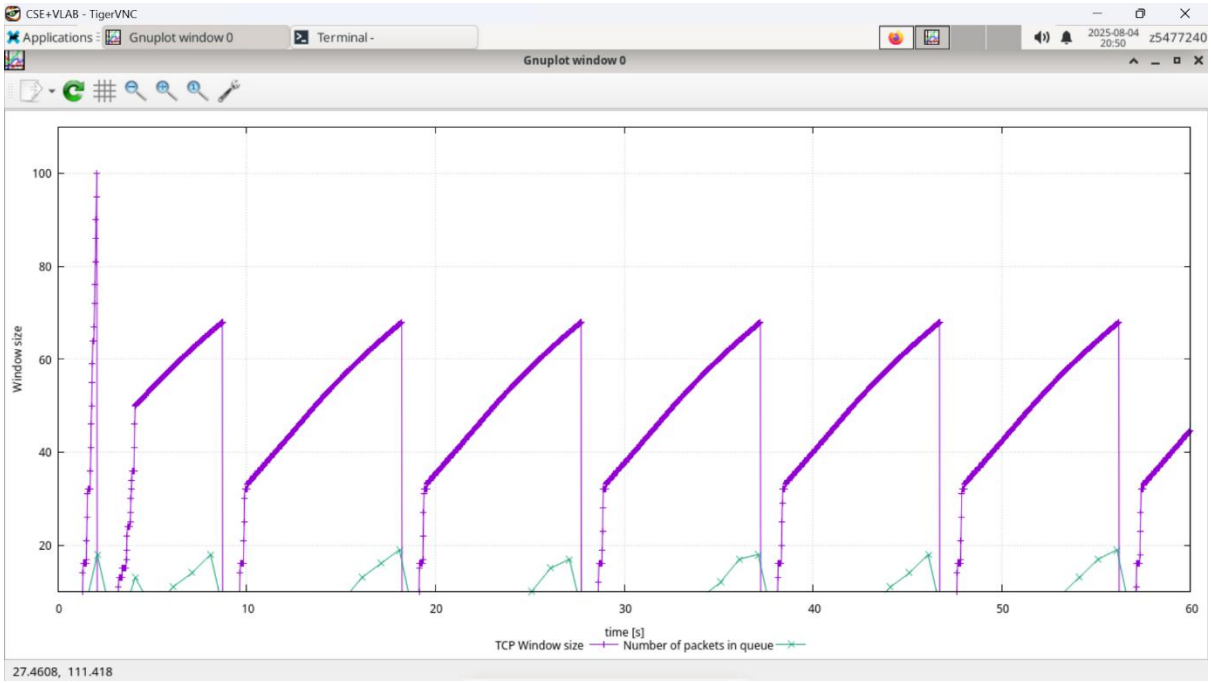
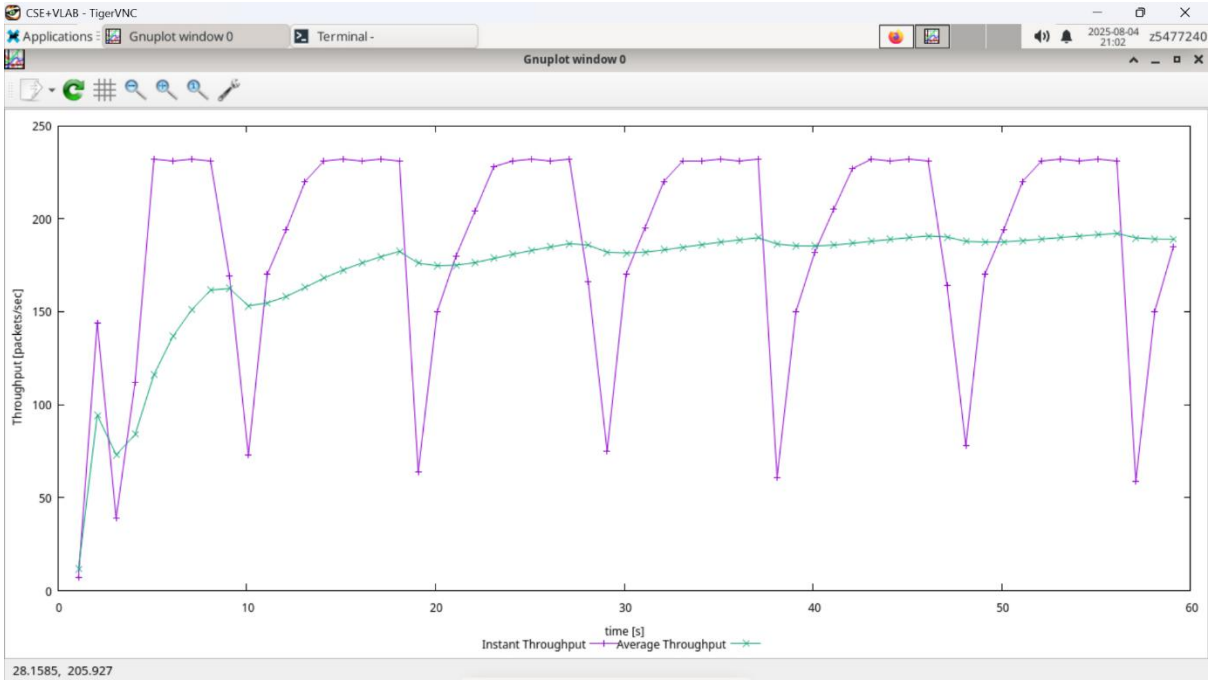
Exercise 1

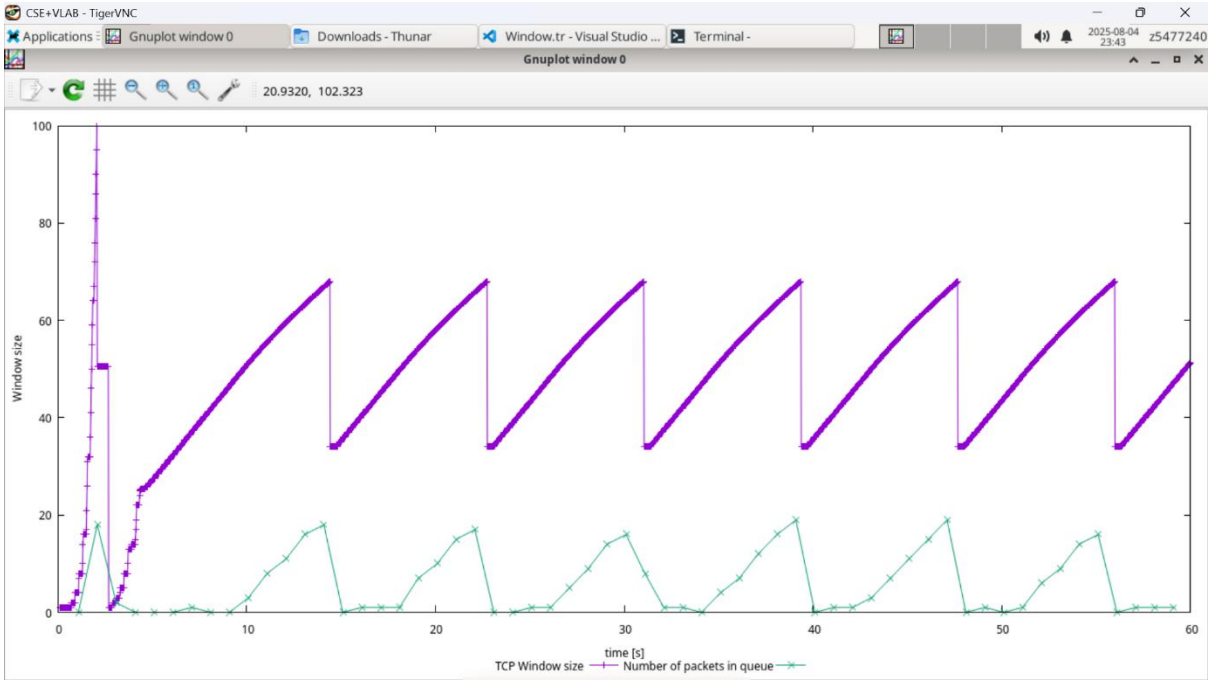


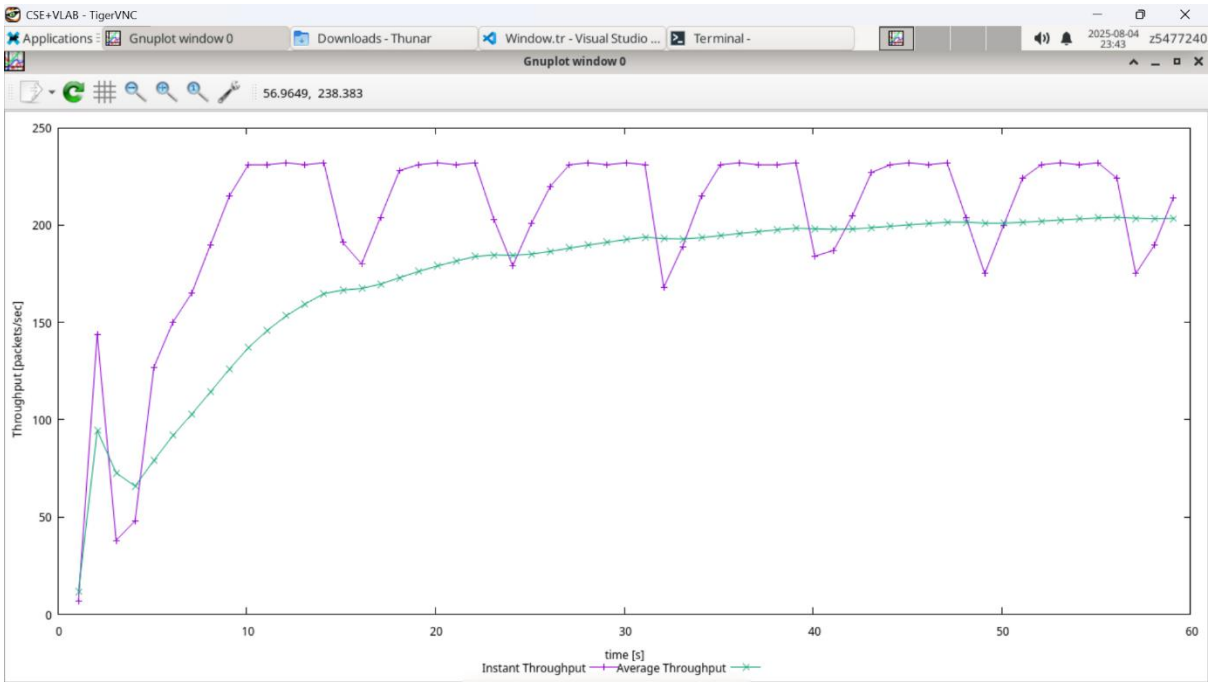
1.

a. The maximum window size is 100.
b. When the window reaches this value, the TCP Tahoe reduces the window size to 1 then starts a slow-start. The slow start. This is to ensure that as little packets are lost due to congestion.
c. The window will grow exponentially until it reaches a certain thresh hold (32 in this diagram). After this it will increase linearly.



2. The average throughput is 188.976 packets/sec which is around 786140bps





3. The average throughput is significantly higher in the Reno graphs and there is less fluctuation and the window size is never set to 0. This is because Reno sets the window size to half rather than 1.

Exercise 2

1. Between 6-8 seconds the throughput is higher for tcp2 since the traffic from n7-n0 directly competes with tcp1.

2. The flow begins with a slow-start (exponential increase) – as the window size increases, this results in the source sending too much leading to packet loss – resulting in the source not sending data and slowly ramps up again causing the fluctuations. The fluctuations decrease as the flow stabilises.

Exercise 3

1. There are two routes: 0-1-4-5 and 2-3-5. These routes do not change over time.
2. The connection between notes 1 and 4 is blocked. The route does not change as a result but rather the packets are dropped.
3. Yes there was additional traffic occurring sporadically. The first route now changes to use 0-1-2-3-5 during 1.0-1.2 seconds.

Questions 4 and 5 are unanswered